Ayush Shrivastava
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Education

- **University of Michigan** — Ann Arbor, MI
  Ph.D. in Computer Science and Engineering — *Aug 2021 – May 2026*
  Advisor: Prof. Andrew Owens
- **Georgia Institute of Technology** — Atlanta, GA
  Master of Science in Computer Science (Specialization: Machine Learning) — *Aug 2019 – May 2021*
  Advisor: Prof. Devi Parikh
- **Indian Institute of Technology (BHU) Varanasi** — Varanasi, India
  Bachelor of Technology in Computer Science and Engineering — *Jul 2014 – May 2018*

Awards

- Received the **PAMI Mark Everingham Prize Career Award (2025)**.

Work Experience

- **Adobe** — Seattle, WA
  Research Scientist | *Manager: Sohrab Amirghodsi* — *Jun 2026 – Present*
  - Working on generative image editing models.

- **University of Michigan** — Ann Arbor, MI
  Graduate Research Assistant | *Supervisor: Prof. Andrew Owens* — *Aug 2021 – May 2026*
  - Worked on problems related to learning correspondences from video, image & video generation. Published papers at *CVPR 2023*, *ECCV 2024*, *CVPR 2025*, *ICLR 2026*.

- **Adobe** — Seattle, WA
  Research Scientist Intern | *Supervisor: Connelly Barnes, Eli Shechtman* — *May 2025 – Mar 2026*
  - Worked on improving the efficiency of video generation models using token pruning techniques. Published at *ICML 2026 F2S Workshop*.

- **Adobe** — Seattle, WA
  Research Scientist Intern | *Supervisor: Connelly Barnes, Eli Shechtman* — *May 2024 – Nov 2024*
  - Built a novel text-to-image diffusion model leveraging camera metadata to precisely control defocus blur while preserving scene content. Published at *WACV 2026*.

- **Amazon Alexa AI** — Remote
  Applied Scientist Intern | *Supervisor: Prof. Jesse Thomason, Robinson Piramuthu* — *May 2021 – Aug 2021*
  - Worked on **TEACh**, a new benchmark released for human–human, interactive dialogues to complete household tasks in simulation on AI2-THOR. Published at *AAAI 2022*.

- **Amazon Alexa AI** Remote
  Applied Scientist Intern | *Supervisor: Dilek Hakkani-Tür* *May 2020 – Aug 2020*
  - Built embodied agents that solve a navigation task by holding dialog in a cooperative setting. Published at *Findings of ACL 2022*.

- **Georgia Institute of Technology** Atlanta, GA
  Graduate Research Assistant | *Supervisor: Prof. Devi Parikh* *Aug 2019 – May 2021*
  - Worked on problems related to Embodied AI and Vision & Language.
  - Worked on using BERT-style models to better ground navigation instructions in 3D environments. Published at *ECCV 2020*.
  - Transferred instruction following agents trained in simulation to robots in real world. Published at *CoRL 2020*.

- **Georgia Institute of Technology** Atlanta, GA
  Visiting Research Scholar | *Supervisor: Prof. Devi Parikh, Prof. Dhruv Batra* *Aug 2018 – May 2019*
  - Worked on solving navigation instruction following in 3D environments by following an ideal agent trajectory. Published at *NeurIPS 2019*.

Publications (* denotes equal contribution)

- **VideoDiffCR: Content-Aware Adaptive Token Pruning for Autoregressive Video Diffusion Transformers** [paper link]
  Ayush Shrivastava, Connelly Barnes, Haoran You, Yifan Gong, Yan Kang, Andrew Owens, Eli Shechtman
  International Conference on Machine Learning (ICML) 2026, From Frames to Stories (F2S) Workshop

- **Point Prompting: Counterfactual Tracking with Video Diffusion Models** [paper link]
  Ayush Shrivastava*, Sanyam Mehta*, Daniel Geng, Andrew Owens
  International Conference on Learning Representations (ICLR) 2026

- **Fine-grained Defocus Blur Control for Generative Image Models** [paper link]
  Ayush Shrivastava, Connelly Barnes, Xuaner Zhang, Lingzhi Zhang, Andrew Owens, Sohrab Amirghodsi, Eli Shechtman
  Winter Conference on Applications of Computer Vision (WACV) 2026, ***Oral***

- **Self-Supervised Spatial Correspondence Across Modalities** [paper link]
  Ayush Shrivastava, Andrew Owens
  Conference on Computer Vision and Pattern Recognition (CVPR) 2025

- **Self-Supervised Any-Point Tracking by Contrastive Random Walks** [paper link]
  Ayush Shrivastava, Andrew Owens
  European Conference on Computer Vision (ECCV) 2024

- **EXIF as Language: Learning Cross-Modal Associations Between Images and Camera Metadata** [paper link]
  Chenhao Zheng, Ayush Shrivastava, Andrew Owens
  Conference on Computer Vision and Pattern Recognition (CVPR) 2023, ***Highlight***

- **TEACh: Task-driven Embodied Agents that Chat** [paper link]
  Aishwarya Padmakumar*, Jesse Thomason*, Ayush Shrivastava, Patrick Lange, Anjali Narayan-Chen, Spandana Gella, Robinson Piramuthu, Gokhan Tür, Dilek Hakkani-Tür
  AAAI Conference on Artificial Intelligence (AAAI) 2022

- **VISITRON: Visual Semantics-Aligned Interactively Trained Object-Navigator** [paper link]
  Ayush Shrivastava, Karthik Gopalakrishnan, Yang Liu, Robinson Piramuthu, Gokhan Tür, Devi Parikh, Dilek Hakkani-Tür
  Findings of the Association for Computational Linguistics (ACL) 2022
  NAACL 2021 Visually Grounded Interaction and Language (ViGIL) Workshop

- **Sim-to-Real Transfer for Vision-and-Language Navigation** [paper link]
  Peter Anderson, Ayush Shrivastava, Joanne Truong, Arjun Majumdar, Devi Parikh, Dhruv Batra, Stefan Lee
  Conference on Robot Learning (CoRL) 2020

- **Improving Vision-and-Language Navigation with Image-Text Pairs from the Web** [paper link]
  Arjun Majumdar, Ayush Shrivastava, Stefan Lee, Peter Anderson, Devi Parikh, Dhruv Batra
  European Conference on Computer Vision (ECCV) 2020, ***Spotlight (Top 5%)***
  ICML 2020 LaReL Workshop, Best Paper Honorable Mention

- **Chasing Ghosts: Instruction Following as Bayesian State Tracking** [paper link]
  Peter Anderson*, Ayush Shrivastava*, Devi Parikh, Dhruv Batra, Stefan Lee
  Neural Information Processing Systems (NeurIPS) 2019

Service Roles and Academic Activities

Reviewer: ICCV [2021], NeurIPS [2021], ICLR [2022, 2024, 2026], CVPR [2023], ACL [2024], ECCV [2024]
Challenge Organization: VQA Challenge [2019, 2020, 2021], Visual Dialog Challenge [2018, 2019, 2020]
Workshop Organization: Visual Question Answering & Dialog Workshop, CVPR [2019, 2020, 2021]